Daeun Kim

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Contact Information

School of Computing, KAIST, 291 Daehak-ro, Yuseong-gu, Daejeon, South Korea, 34141

Research Interests

- Computer architecture
- Generative AI system
- HW/SW Co-design

Education

KAIST, Ph.D. in School of computing — Mar. 2025 – date
- Advisor: Prof. Jongse Park

KAIST, M.S. in School of computing — Mar. 2023 – Feb. 2025
- Advisor: Prof. Jongse Park
- Thesis: Exploiting MX Quantization for Accelerating Diffusion Transformer-based Image Generative Model Inference

KAIST, B.S. in School of computing — Mar. 2018 – Feb. 2023

Publications

Understanding and Exploiting Diagonal Attention Sparsity in Autoregressive Image Generation — Sep. 2026

Daeun Kim, Junwha Hong, Changhun Oh, Yoonsung Kim, Yoonhyeong Lee, Jongse Park

IISWC 2026

Déjà Vu: Efficient Video-Language Query Engine with Learning-based Inter-Frame Computation Reuse — Sep. 2025

Jinwoo Hwang, ***Daeun Kim***, Sangyeop Lee, Yoonsung Kim, Guseul Heo, Hojoon Kim, Yunseok Jeong, Tadiwos Meaza, Eunhyeok Park, Jeongseob Ahn, Jongse Park

VLDB 2025

MixDiT: Accelerating Image Diffusion Transformer Inference with Mixed-Precision MX Quantization — Apr. 2025

Daeun Kim, Jinwoo Hwang, Changhun Oh, Jongse Park

IEEE Computer Architecture Letters (CAL)

CoVA: Exploiting Compressed-Domain Analysis to Accelerate Video Analytics — Jun. 2022

Jinwoo Hwang, Minsu Kim, ***Daeun Kim***, Seungho Nam, Yoonsung Kim, Dohee Kim, Hardik Sharma, Jongse Park

USENIX ATC 2022

Honors & Experiences

VLDB student grant — Sep. 2025

ISCA student grant — Jun. 2023

Undergrad Architecture Mentoring (uArch) Workshop mentee, ISCA 2023 — Jun. 2023

Undergraduate research intern, KAIST CASys Lab — Sep. 2021 - Feb. 2023

AI engineer internship, Naviworks Co.,Ltd. — Mar. 2021 - Aug. 2021

Teaching Assistant

KAIST CS50100: Computer Architecture — Spring 2026
KAIST CS230: System Programming — Fall 2022

References

Jongse Park jspark@casys.kaist.ac.kr
Professor, KAIST